

Practical No. 9

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Aim : Create Decision Table for a Project

1. Aim

To prepare a decision table for the Smart Education Platform that shows how different conditions lead to different actions when a learner enrolls in a course.

2. Objective

To represent enrollment business rules in a structured form and identify the actions that the system should perform for different combinations of conditions.

3. Project Scenario

In a Smart Education Platform, a learner selects a course and requests enrollment. The system checks whether enrollment for the course is open, whether the course has available seats, and whether any required payment is successful. Based on these conditions, the learner is enrolled, asked to retry payment, or informed that enrollment cannot proceed.

4. Conditions and Actions

Type	Code	Condition / Action
Condition	C1	Enrollment for the selected course is open?
Condition	C2	Seats are available in the course?
Condition	C3	Required payment is successful?
Action	A1	Confirm enrollment and provide course access
Action	A2	Show payment retry / payment failure message
Action	A3	Reject enrollment and inform the learner

5. Decision Table

Conditions / Actions	R1	R2	R3	R4	R5
C1: Enrollment open?	Y	Y	Y	Y	N
C2: Seats available?	Y	Y	N	Y	-
C3: Required payment successful?	Y	N	-	Y	-
A1: Confirm enrollment and provide access	✓	✗	✗	✓	✗
A2: Show payment retry / failure	✗	✓	✗	✗	✗
A3: Reject enrollment and inform learner	✗	✗	✓	✗	✓

6. Explanation of Rules

- Rule 1 (R1): Enrollment is open, seats are available, and required payment succeeds (or no payment is required) → the system confirms enrollment and provides course access.
- Rule 2 (R2): Enrollment is open and seats are available, but required payment fails → the system asks the learner to retry or select another payment method. Enrollment remains pending.
- Rule 3 (R3): Enrollment is open, but no seats are available → the system rejects the enrollment request and informs the learner that the course is full.
- Rule 4 (R4): Enrollment is open, seats are available, and no payment is required → the system confirms enrollment and provides course access.
- Rule 5 (R5): Enrollment for the selected course is closed → the system rejects the enrollment request regardless of seat availability or payment status.

7. Sample Decision Table with Real-Life Example

Situation	System Decision
Enrollment open + seat available + payment successful	Enrollment Confirmed; Course Access Provided
Enrollment open + seat available + payment failed	Retry Payment; Enrollment Pending
Enrollment open + course full	Enrollment Rejected / Course Full
Enrollment closed	Enrollment Rejected / Enrollment Closed
Enrollment open + seat available + course is free	Enrollment Confirmed; Course Access Provided

8. Result / Conclusion

The decision table for the Smart Education Platform was successfully prepared. It represents different combinations of enrollment conditions and the corresponding system actions. Decision tables help software developers and testers understand business rules clearly, identify missing cases, and design test cases for different situations.